IEEE Returning Mothers Conference 2025
IDEATHON AND PITCHFEST

Unleash Ideas. Empower Innovation. Inspire Comebacks.

SUBMISSION FORM

1. Project Summary

Include the following in your summary:

A. Problem Statement (50 words) - Clearly define the issue you're addressing.

India’s heritage sites face rapid deterioration, restricted access, and high maintenance costs, limiting cultural preservation and global reach. Tourists and students often miss authentic experiences due to seasonal festivals or inaccessible areas. There is a need for an immersive, sustainable, and accessible solution to preserve and showcase India’s cultural heritage.

B. Proposed Solution (200 words) - Describe how your idea tackles the problem. Include any technology, process, or social impact strategy.

To address the challenges of deteriorating heritage sites, limited accessibility, and high upkeep costs, we propose a digital preservation and immersive showcasing platform powered by cutting-edge technologies. Using **Reality Capture** and **Blender**, heritage sites and monuments will be digitally restored into highly detailed 3D models without losing originality. These assets will then be transformed into interactive VR environments through **Unreal Engine and Unity**, creating lifelike, immersive experiences that preserve cultural authenticity.

Our platform will be accessible through three modes: **website, mobile app, and VR kiosks**. The website will serve as a cultural hub, promoting India's artisans and handicrafts, enabling tourists to purchase authentic products, and providing educational resources such as AI-powered historian chatbots and cultural archives. The **AR mobile app** will guide tourists and students with interactive cultural maps, navigation support, and site-specific historical information. The **VR solution** will be deployed at airports, museums, and public spaces through booths and kiosks, allowing users to explore restricted heritage zones and virtually experience local festivals anytime, even beyond seasonal availability.

This multi-platform approach not only preserves India's cultural heritage in a sustainable digital format but also boosts tourism, empowers artisans, enhances learning for students, and fosters pride and global recognition of India's traditions.

C. Impact Description (300 words) - How does your solution impact specific metrics (e.g., health, environment, economy)? Highlight its contribution to the SDGs

Our solution directly impacts cultural preservation, education, tourism, and the economy by offering a sustainable and immersive way to safeguard and showcase India's heritage. By digitally restoring monuments and traditions, we prevent knowledge loss and create a permanent archive for future generations, addressing the urgent challenge of deterioration and limited physical access.

Economic Impact: The platform promotes local artisans by providing global access to their handicrafts through e-commerce integration. This creates new revenue streams for rural communities, fosters entrepreneurship, and strengthens India's cultural economy. Tourism also benefits, as immersive AR and VR experiences attract more visitors, generating higher footfall in museums, heritage sites, and cultural hubs.

Educational Impact: Students, researchers, and historians gain an interactive learning tool that goes beyond textbooks. By engaging with VR museums, AR navigation, and historian chatbots, learners access authentic knowledge in an engaging format. This promotes inclusive and equitable quality education, aligning with **SDG 4 (Quality Education)**.

Cultural & Social Impact: Citizens and tourists can virtually experience cultural festivals, restricted heritage zones, and traditional practices, increasing cultural pride and awareness. This contributes to **SDG 11 (Sustainable Cities and Communities)** by preserving cultural heritage while ensuring inclusive access.

Environmental Impact: By digitizing heritage, we reduce the physical strain on fragile monuments caused by excessive tourism, supporting **SDG 13 (Climate Action)** and **SDG 15 (Life on Land)** through sustainable preservation.

Overall, our solution bridges technology with tradition, creating measurable outcomes: increased artisan income, higher tourism engagement, enriched student learning, and long-term preservation of cultural assets. By promoting accessibility, sustainability, and inclusivity, it aligns with India's vision of using technology for social good and contributes to multiple Sustainable Development Goals, ensuring that India's cultural richness continues to inspire globally.

2. Goal Definition

A. Primary Goal/Metric: What key metric are you targeting (e.g., reduction in emissions, increase in access, improvement in efficiency)?

The primary goal of our project is to significantly increase access to India's cultural heritage by leveraging digital technologies. We aim to digitally preserve monuments, traditions, and festivals and make them accessible to a broader audience through web platforms, AR applications, and VR kiosks. Our key metric is to increase accessibility to heritage sites and cultural experiences by at least **70%**, especially for students, tourists, and researchers who face limitations due to distance, time, or restricted site access. This metric ensures inclusive engagement while also fostering cultural pride, tourism growth, and sustainable preservation of fragile monuments.

B. Differentiator: How does your solution differ from existing ones? What's your innovation edge?

Our solution stands apart from existing digital heritage initiatives by combining **multi-platform accessibility, cultural commerce, and immersive festival experiences** into a single ecosystem. While many projects focus only on static digital archives or 360° virtual tours, we go further by using **Reality Capture, Blender, and Unreal/Unity** to create lifelike, interactive 3D environments that preserve both the architectural and cultural essence of heritage sites. Unlike traditional solutions, our platform integrates **AI-powered historian chatbots**, enabling dynamic knowledge-sharing and personalized learning experiences.

Additionally, our approach emphasizes **economic empowerment** by connecting artisans and handicraft makers directly to global markets through the website's cultural e-commerce platform, creating sustainable livelihoods. The inclusion of **AR mobile guidance** helps tourists and students navigate heritage locations while learning contextually, bridging physical visits with digital enrichment. Our **VR kiosks and booths** at airports, museums, and public spaces provide immersive access to restricted heritage areas and off-season festivals, ensuring no cultural experience is missed due to time or location constraints.

This holistic integration of **digital preservation, immersive education, cultural commerce, and tourism enhancement** makes our project innovative, scalable, and impactful, offering a future-ready model to preserve and promote India's cultural heritage globally.

C. Current Development Status: Where do you stand currently—ideation, prototyping, pilot testing, implementation, etc.?

Our project has advanced well beyond the ideation stage and is currently at the **prototyping and early pilot testing phase**. We have successfully obtained permission from the **Archaeological Survey of India (ASI)** and captured the intricate **pillars of the Sri Dhenupureshwarar Temple** using Reality Capture technology. These models have been digitally refined in Blender and integrated into immersive environments using **Unreal Engine and Unity**, resulting in a functional **VR prototype** that allows users to explore the temple virtually.

In parallel, we have developed **AR mobile applications** that provide navigation support, cultural mapping, and contextual information for tourists and students. A **dedicated website** has also been created, serving as a cultural hub where users can access heritage content, learn about traditions, and explore artisan products. Together, these components demonstrate the feasibility of our multi-platform approach.

Currently, we are testing these prototypes with a small group of users to collect feedback on usability, immersion, and cultural authenticity. The next steps involve expanding the scope to additional monuments, setting up **VR kiosks at public spaces and museums**, and scaling our AR and web solutions for broader access.

3. Implementation Plan

A. Location of Execution: (Global, National, Regional, or specify)

The project is planned for implementation at the **national level across India**, covering heritage sites, monuments, and cultural hubs across multiple states. Starting with selected pilot sites such as the Sri Dhenupureshwarar Temple, the platform will gradually expand to include diverse monuments, museums, and cultural festivals across the country. The aim is to create a **comprehensive digital archive of India's heritage**, accessible globally through web, AR, and VR platforms, while ensuring that the primary focus remains on **nationwide preservation, promotion, and accessibility** of India's rich cultural legacy.

B. Resources Required: Briefly mention technical, financial, or human resources required. If applicable, describe how this aligns with sustainable and inclusive practices

The successful execution of this project requires a blend of **technical, financial, and human resources**. On the technical side, we need **Reality Capture equipment (high-resolution cameras, drones, LiDAR scanners)**, powerful computing systems for 3D modeling and rendering, and software tools such as **Blender, Unreal Engine, and Unity**. For deployment, we require infrastructure to host the **website, AR mobile applications, and VR kiosks**, supported by cloud storage for scalability.

Financially, resources will be allocated for procuring capture equipment, software licenses, cloud hosting, and VR/AR hardware such as headsets and kiosks. Funding will also cover partnerships with artisans and marketing efforts to promote cultural commerce and tourism.

In terms of human resources, the project involves a multidisciplinary team: **heritage experts, archaeologists, 3D artists, software developers, AI specialists, designers, and tourism consultants**. Collaborations with cultural institutions and the Archaeological Survey of India (ASI) ensure accuracy and compliance.

Aligned with **sustainable and inclusive practices**, the project minimizes physical strain on monuments by shifting experiences to digital platforms, empowers rural artisans by connecting them to global buyers, and provides equitable access for students and citizens nationwide.

C. Business Model/Scalability Plan (Optional): Describe how you plan to scale this idea.

Our project is designed with a **hybrid business model** to ensure both accessibility and long-term sustainability. Initially, we will provide **free access** to selected heritage sites and cultural experiences through our website and mobile app to attract students, researchers, and tourists. As the platform grows, we will introduce **premium content** such as full VR environments, access to restricted heritage areas, and virtual festivals, available through **subscription plans or one-time purchases**. For institutions like schools, colleges, and museums, we will offer **customized enterprise packages** for cultural education and training.

To support artisans and local communities, the integrated **cultural e-commerce platform** allows tourists to purchase handicrafts directly, generating revenue for artisans while enabling us to earn a **small commission**. Partnerships with government bodies, tourism boards, and private sponsors will further sustain development. VR kiosks installed at **airports, museums, and cultural hubs** will operate on a **pay-per-use or rental model**, creating an additional income stream.

Scalability is built into the model by digitizing heritage sites across different states in phases, eventually covering the **entire nation**. The digital archive will also be accessible globally, expanding tourism outreach, promoting India's cultural richness worldwide, and making the platform financially viable and socially impactful.

4. Awareness and Ecosystem Fit

A. Do you know of any relevant schemes (Govt/Non-Govt) that support such ideas?

Yes, several schemes support such initiatives. The **Ministry of Culture's "Digital India" mission** promotes digital preservation of heritage. The **Adopt a Heritage Scheme** encourages public-private partnerships for site development. Additionally, **Incredible India 2.0** focuses on immersive tourism experiences, while **Startup India** and **Atal Innovation Mission (AIM)** provide funding and incubation support for technology-driven cultural projects.

B. Likelihood of Success: (Rate on a scale of 1–10 and justify briefly)

Likelihood of Success: 9/10
The project has a very high likelihood of success because it leverages **proven technologies** like Reality Capture, Blender, Unreal/Unity, AR, and VR, which are already widely used in cultural preservation and training. With support from government bodies like the **Archaeological Survey of India (ASI)** and alignment with national initiatives such as **Digital India** and **Incredible India**, the ecosystem is favorable. Additionally, the project creates value across multiple sectors—**education, tourism, economy, and cultural preservation**—making it scalable and sustainable. The only challenge lies in securing consistent funding and wide adoption, which can be overcome through partnerships.

C. Areas where mentorship is needed: (Select multiple if needed)

1. Market Research
2. Business Model
3. Team Building
4. Funds Management
5. Resource Acquisition
6. Technical Support
7. Scaling

Business Model – to design a sustainable revenue strategy combining free access, premium content, and cultural e-commerce.
Funds Management – to efficiently allocate resources for technology, partnerships, and scaling.
Scaling – to expand from pilot projects to a nationwide platform while maintaining cultural authenticity and inclusivity.

Let us innovate with purpose
Aligned with global goals for a better future.